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Different Types of Matrix

Types of Matrices:

TRANSPOSE OF A MATRIX

Matrix Transpose: Transpose of a matrix A is a matrix obtained by interchanging the rows and columns of A.

Denoted by A^T or A'.

Example:

$$\underset{4x1}{X} = \begin{bmatrix} 12 \\ 9 \\ -4 \\ 0 \end{bmatrix} \quad \underset{1x4}{X'} = \begin{bmatrix} 12 & 9 & -4 & 0 \end{bmatrix}$$

$$\underset{3x4}{A} = \begin{bmatrix} 21 & 62 & 33 & 93 \\ 44 & 95 & 66 & 13 \\ 77 & 38 & 79 & 33 \end{bmatrix} \quad \underset{4x3}{A'} = \begin{bmatrix} 21 & 44 & 77 \\ 62 & 95 & 38 \\ 33 & 66 & 79 \\ 93 & 13 & 33 \end{bmatrix}$$

Symmetric Matrix: A symmetric matrix is a square matrix that is equal to its transpose. Formally, A will be symmetric if and only if $A = A^t$. For example,

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{bmatrix}_{3\times3} ; A^t = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{bmatrix}_{3\times3} ; \text{ Hence, } A = A^t.$$

Because equal matrices have equal dimensions, only square matrices can be symmetric.

Basic properties

- The sum and difference of two symmetric matrices is symmetric.
- This is not always true for the product: given symmetric matrices A and B, then AB is symmetric if and only if A and B commute, i.e., if AB=BA.
- For any integer n, A^n is symmetric if A is symmetric.

- If A^{-1} exists, it is symmetric if and only if A is symmetric.
- Rank of a symmetric matrix A is equal to the number of non-zero eigenvalues of A.

SYMMETRIC MATRIX

Construction:

1. write the principal diagonal with any entries ;
2. complete 1st row with any entries ;
3. complete 1st column with those entries used in 2;
4. complete 2nd row with any entries ;
5. complete 2nd column with those entries used in 4;
6. complete other rows & columns using the same process.

- A 4×4 example:

$$M = \begin{bmatrix} 2 & 5 & 6 & -7 \\ 5 & -3 & 2 & -1 \\ 6 & 2 & 0 & 4 \\ -7 & -1 & 4 & 5/2 \end{bmatrix}$$

Skew-Symmetric Matrix: A skew-symmetric matrix is a square matrix whose transpose equals its negative. Formally, A will be skew-symmetric if and only if $A = -A^t$. For example,

$$A = \begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & -5 \\ 4 & 5 & 0 \end{bmatrix}_{3\times3} ; A^t = -\begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & -5 \\ 4 & 5 & 0 \end{bmatrix}_{3\times3} ; \text{ Hence, } A = -A^t.$$

Because equal matrices have equal dimensions, only square matrices can be skew-symmetric.

SKEW-SYMMETRIC MATRIX

Construction:

1. write the principal diagonal with 0's;
2. complete the 1st row with any entries ;
3. complete the 1st column with -ve entries of 2 ;
4. complete the 2nd row with any entries ;
5. complete the 2nd column with -ve entries of 4 ;
6. complete other rows & columns using the same process.

- A 4×4 example:

$$M = \begin{bmatrix} 0 & -2 & 6 & -9 \\ 2 & 0 & 3 & 7 \\ -6 & -3 & 0 & 0 \\ 9 & -7 & 0 & 0 \end{bmatrix}$$

THEOREM 1

For any square matrix A with real number entries,

a. $A + A^T$ is a symmetric matrix and

b. $A - A^T$ is a skew symmetric matrix.

Proof for symmetric:

Let $B = A + A^T$, then

$B^T = (A + A^T)^T$

$= A^T + (A^T)^T \quad (\because (A+B)^T = A^T + B^T)$

$= A^T + A \quad (\because (A^T)^T = A)$

$= A + A^T$

$B^T = B$

$B = A + A^T$ is a symmetric matrix

Proof for skew-symmetric:

Let $C = A - A^T$, then

$C^T = (A - A^T)^T$

$= A^T - (A^T)^T \quad (\because (A-B)^T = A^T - B^T)$

$= A^T - A \quad (\because (A^T)^T = A)$

$= -(A - A^T)$

$C^T = -C$ which means $C = -C^T$

$C = A - A^T$ is a skew-symmetric matrix

THEOREM 2

Any square matrix can be expressed as the sum of a symmetric and a skew-symmetric matrix.

Proof:

Let A be a square matrix represented as

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{Symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{Skwe-Symmetric}}$$

(Proved in theorem 1)

Also, since $(kA)^T = kA^T$ ... It means $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$ is skew-symmetric.

HOW TO APPLY THEOREM 2

For symmetric part	For skew-symmetric part
Write M^T	Write M^T
Find $M + M^T$	Find $M - M^T$
Calculate $\frac{1}{2}(M + M^T)$	Calculate $\frac{1}{2}(M - M^T)$

Now M = sym + skew

EXPRESS $M = \begin{pmatrix} 1/3 & -5 \\ 0 & 8 \end{pmatrix}$ AS THE SUM OF A SYMMETRIC & A SKEW-SYMMETRIC MATRIX.

Solution: Given, $M = \begin{pmatrix} 1/3 & -5 \\ 0 & 8 \end{pmatrix}$, $M^T = \begin{pmatrix} 1/3 & 0 \\ -5 & 8 \end{pmatrix}$

- symmetric part

$$M + M^T = \begin{pmatrix} \frac{1}{3} & -5 \\ 0 & 8 \end{pmatrix} + \begin{pmatrix} \frac{1}{3} & 0 \\ -5 & 8 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{3} & -5 \\ -5 & 16 \end{pmatrix}$$

$$\therefore ½(M + M^T) = \begin{pmatrix} \frac{1}{3} & -\frac{5}{2} \\ -\frac{5}{2} & 8 \end{pmatrix}$$

- skew-symmetric part

$$M - M^T = \begin{pmatrix} \frac{1}{3} & -5 \\ 0 & 8 \end{pmatrix} - \begin{pmatrix} \frac{1}{3} & 0 \\ -5 & 8 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -5 \\ 5 & 0 \end{pmatrix}$$

$$\therefore ½(M - M^T) = \begin{pmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{pmatrix}$$

Now $M = \begin{pmatrix} \frac{1}{3} & -\frac{5}{2} \\ -\frac{5}{2} & 8 \end{pmatrix} + \begin{pmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{pmatrix}$

Hermitian Matrix: A Hermitian matrix (or self-adjoint matrix) is a complex square matrix that is equal to its own conjugate transpose- that is, the element in the i-th row and j-th column is equal to the complex conjugate of the element in the j-th row and i-th column, for all indices i and j:

$a_{ij} = \overline{a_{ji}}$ in matrix form $A = \overline{A^T}$.

Example: $A = \begin{bmatrix} 3 & 1+i \\ 1-i & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2+i & 5-4i \\ 2-i & 4 & 6i \\ 5+4i & -6i & 2 \end{bmatrix}$

The general form of skew-Hermitian matrix is,

$$\begin{bmatrix} a & b+ci \\ b-ci & d \end{bmatrix}$$

Skew-Hermitian Matrix: A square matrix with complex entries is said to be skew-Hermitian or anti-Hermitian if its conjugate transpose is the negative of the original matrix. That is, the matrix A is skew-Hermitian if it satisfies the relation,

$$A^{H} = -A$$

where A^{H} denotes the conjugate transpose of the matrix A. In component form, this means that

$$a_{ij} = -\overline{a_{ji}}$$

for all indices *i* and *j*, where a_{ij} is the element in the *i*-th row and j-th column of A, and the overline denotes complex conjugation.

Example: $B = \begin{bmatrix} 0 & 2+i & 5+4i \\ -2+i & 4i & 1-6i \\ -5+4i & -1-6i & -2i \end{bmatrix}$

The general form of skew-Hermitian matrix is,

$$\begin{bmatrix} ai & b+ci \\ -b+ci & di \end{bmatrix}$$

Problem 01: Prove that, $A = \begin{bmatrix} 3i & 1+i \\ -1+i & -i \end{bmatrix}$ is a skew-Hermitian matrix.

Solution:

Let us see how, $A^{T} = \text{Transpose of } A = \begin{bmatrix} 3i & -1+i \\ 1+i & -i \end{bmatrix}$

$$A^H = \text{Conjugate transpose of } A = \begin{bmatrix} -3i & -1-i \\ 1-i & i \end{bmatrix}$$

$$\text{Again, } -A = \begin{bmatrix} -3i & -1-i \\ 1-i & i \end{bmatrix}$$

Thus $A^H = -A$. Hence, A is a skew Hermitian matrix.

Orthogonal Matrix: The matrix A is an orthogonal matrix or orthonormal matrix if and only

$$AA^T = A^T A = I$$

Where A^T is the transpose of A and I is the identity matrix.

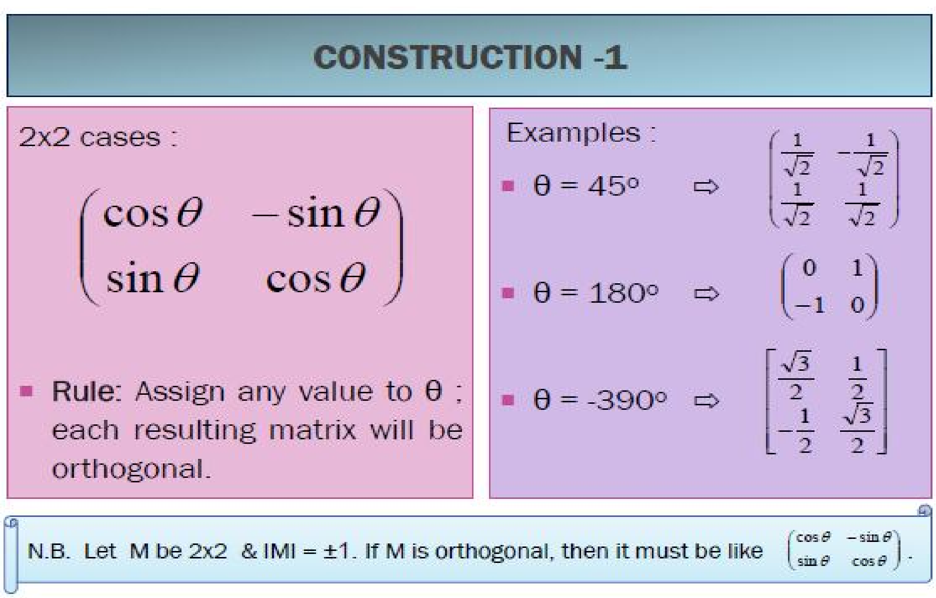
This leads to the equivalent characterization: a matrix Q is orthogonal if its transpose is equal to its inverse:

$$A^T = A^{-1}$$

where Q^{-1} is the inverse of Q.

Example:

$$\begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$$



CONSTRUCTION -2

- 3x3 Cases :

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & \pm 1 \end{bmatrix} \quad \forall\, \theta$$

- 4x4, 5x5, ... cases are made in the same way, i.e. proceed with ±1 along the diagonal. Then complete the rest of rows & columns with zeroes only.
- Not all 3x3, 4x4, ... orthogonal matrices conform to this pattern.

Examples :

- A 3x3 orthogonal matrix:

(taking $\theta = 120°$ we get)

$$\begin{pmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- A 4x4 orthogonal matrix:

$$\theta = \pi \;\Rightarrow\; \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Important Formulas for Matrices:

If A, B are square matrices of order n, and I_n is a corresponding unit matrix, then

a. A(adj.A) = | A | I_n = (adj A) A

b. adj (AB) = (adj B) (adj A)

c. adj (A^m) = $(\text{adj } A)^m$,

d. A is symmetric ⇒adj A is also symmetric

e. A is diagonal ⇒adj A is also diagonal

f. A is triangular ⇒adj A is also triangular

g. A is singular ⇒| adj A | = 0

Exercise

Q-1. Write as the sum of a symmetric & a skew symmetric matrix where :

❶ $A = \begin{pmatrix} 0 & 7 \\ \frac{4}{9} & -5 \end{pmatrix}$; ❷ $B = \begin{pmatrix} 3 & \frac{6}{7} & 5 \\ -5 & 0 & \frac{1}{2} \\ 0 & 4 & -8 \end{pmatrix}$; ❸ $C = \begin{pmatrix} 3 & 7 & \frac{3}{11} & -9 \\ 9 & 6 & 0 & \frac{5}{6} \\ -7 & 4 & -5 & 2 \\ -1 & 0 & 3 & 0 \end{pmatrix}$

Q-2. Give 3 examples of symmetric matrices: (i). 4x4 ; (ii). 5x5 ; (iii). 6x6

Q-3. Give 3 examples of skew- symmetric matrices: (i). 4x4 ; (ii). 5x5 ; (iii). 6x6

1. Which matrices can be expressed as the sum of a symmetric & skew-symmetric matrix?
2. What is the trace of a 29x29 matrix : (i) skew-symmetric matrix ; (ii) symmetric matrix ?
3. Give an example of 7x7 matrix which is both: symmetric and skew-symmetric matrix.
4. A 99×99 skew-symmetric matrix was found to have the trace "0": **True or, False**?